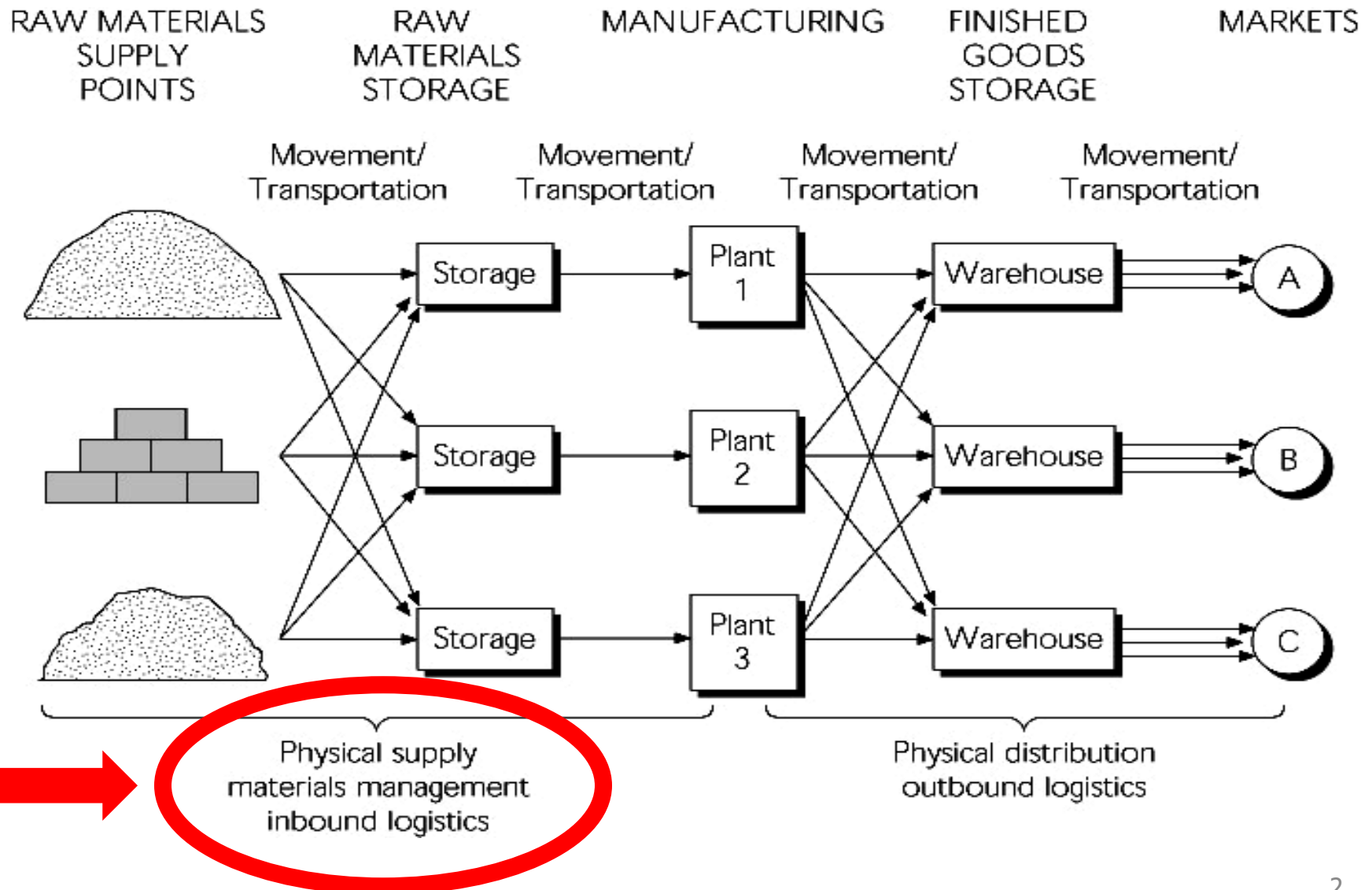


Section 2

Material Management

Material Management



Material Management

- **Definition**

Materials management is the planning and control of the flow of materials

- **Materials Management Activities**

- Purchasing/Procurement
- Inventory Management
- Inbound Transportation (Logistics)
- Formalities
- Etc.

Purchasing

- The process of obtaining goods and services from outside sources
- The best quality and prompt delivery
- Consist of
 - Procurement (pre-award processes)
 - Contract Administration (post-award processes)

Purchasing Roles

Purchasing have legitimate authority:

- Prepare Tender Plan
- Evaluate and Select Suppliers
- Negotiate with Suppliers
- Review Specifications
- The Primary Contact with Suppliers
- Awarding Purchase Orders and Contracts
- Determine Type of Contract

Purchasing “Musts”

- Right “stuff”
- Right price
- Right quality
- Right quantity
- Right time
- Right place
- Right source

Purchasing Process

PREAWARD PROCESSES (PROCUREMENT)

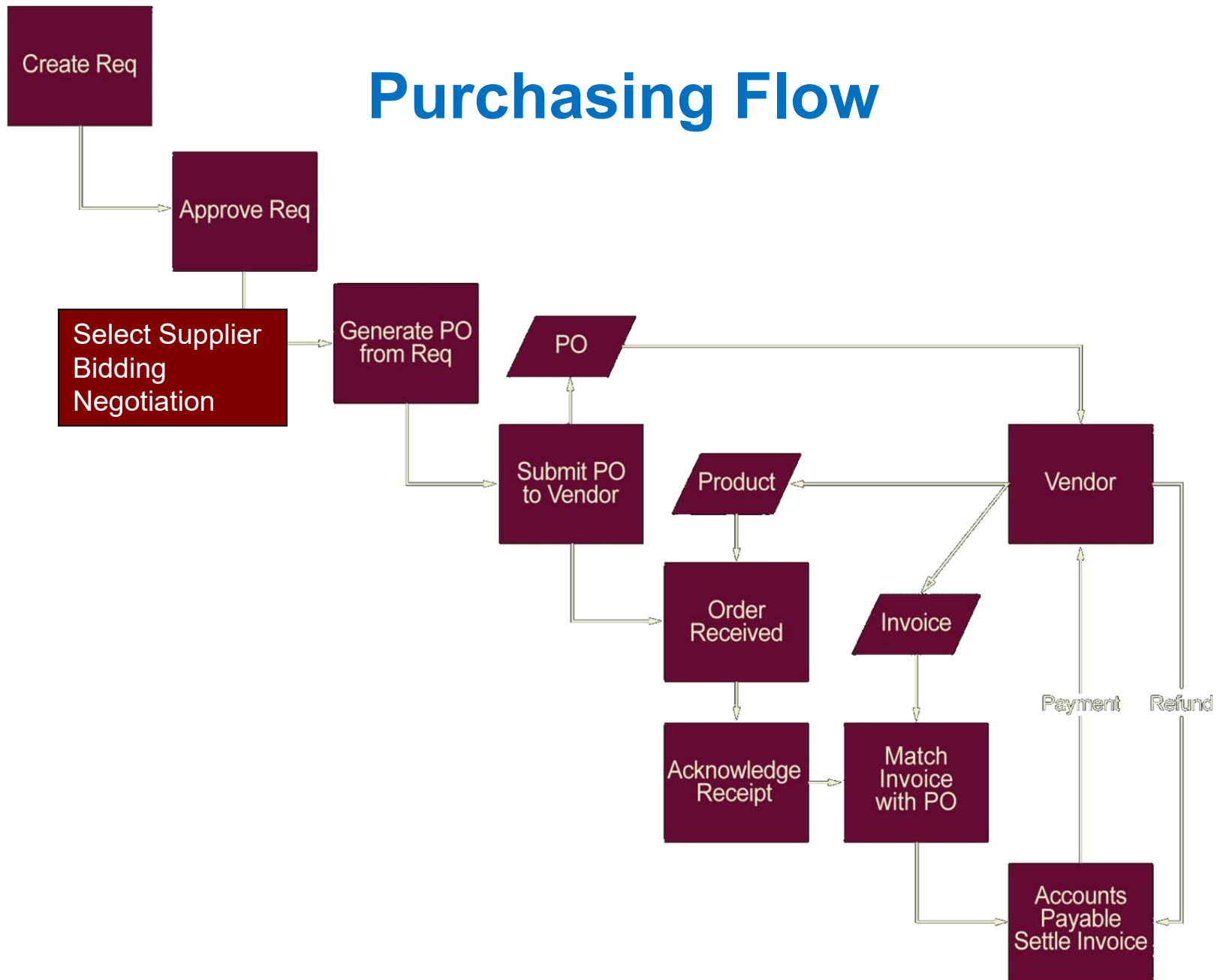
1. Recognition of need or Identification requirements
2. Description or SOW review
3. Selection sources of supply
4. Solicitation (oral or written)
5. Determination/negotiation of price and terms
6. Preparation and placement of the purchase order/contract

Purchasing Process

POSTAWARD (CONTRACT ADMINISTRATION) PROCESSES

7. Monitoring and/or expedite the order
8. Receipt, inspection, and acceptance of goods
9. Process the invoice and pay the supplier
10. Maintenance of records

Purchasing Flow

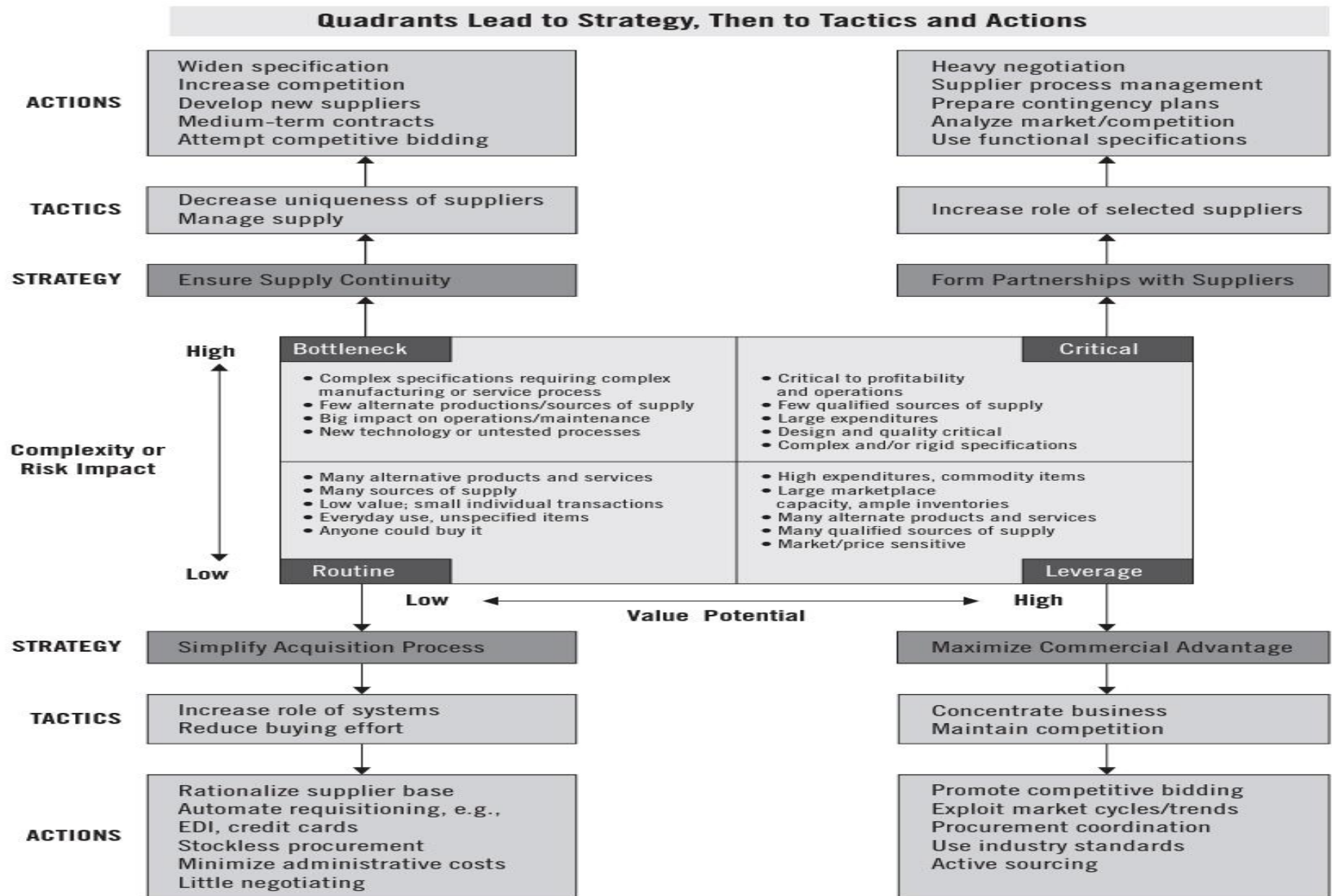


Purchasing Strategic

Purchasing Strategic:

- Supply Base Optimization
- Developing Commodity Strategic Sourcing
- Strategic Cost Analysis
- EPI & ESI (Supplier Integration)
- Negotiation
- Market Research
- Global Sourcing
- **Outsourcing (IPM)**

Developing Commodity Strategic Sourcing



Contract Administration

Contract Administration is any activity taken by the Company during the time from contract award to contract closeout

- Assuring buying firm fulfills its part of the contract agreement
- Making sure that the supplier's performance complies with the contract

Contract Administration Planning

Contract Administration must answer:

- What must be done
- What tasks must be performed
- Who must accomplish the what?
- Where will contract activities take place
- When must each activity take place
- How will tasks be accomplished

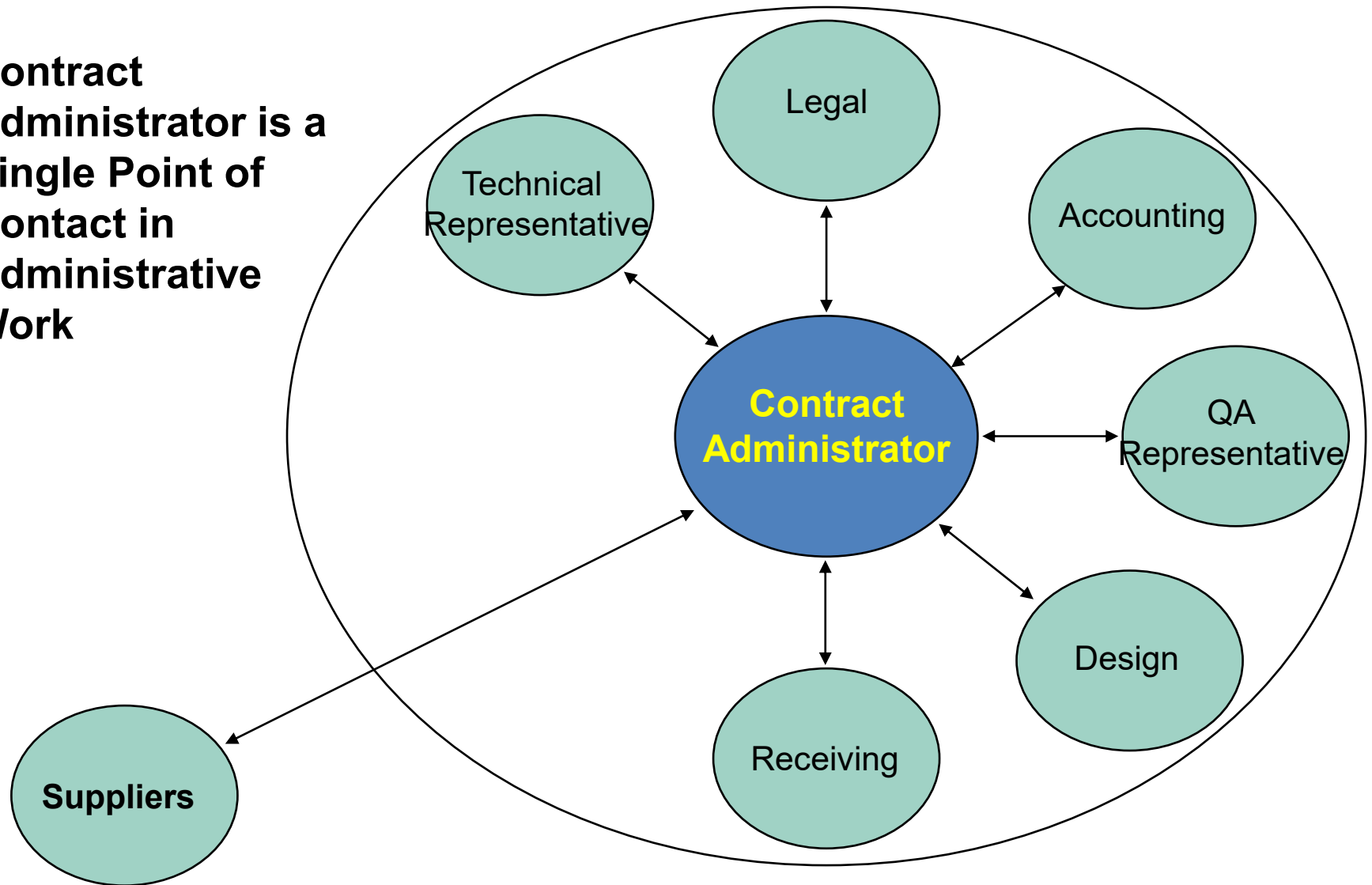
Contract Administration Tasks

Contract Administration Plans must address all the phases of contract administration:

- Contract performance preliminaries
- Contract performance monitoring
- Inspection and acceptance
- Post contract activity
- Contract closeout

The Contract Administration Team

Contract Administrator is a Single Point of Contact in Administrative Work



Contract Closeout

- Contract closeout is the final step where both parties confirm that the work has been completed, and accepted, and all contract paperwork is finished
- The contracting administrator should sign a contract completion statement confirming that all administrative actions have been performed.

Contract Closeout

- Lower dollar-value contracts and purchase orders should be considered closed with evidence of receipt
- Other than lower dollar-value purchase orders and contracts should be closed out

Closeout Steps

1. Verify that the contract is physically complete
2. Obtain all forms, reports, and clearances
3. Verify that all other terms and conditions have been met
4. Settle any outstanding issues

Closeout Steps

5. Verify no outstanding claims or disputes
6. Determine whether to invoke phase-in/phase-out rights
7. Make final payment
8. Prepare contract and completion statement
9. Dispose of files

Contract Completion Statement

CONTENT REQUIREMENTS OF A CONTRACT COMPLETION STATEMENT

1. Contracting office name and address.
2. Contract number.
3. The last modification number.
4. Last call or order number.
5. The supplier's name and address.
6. Dollar amount of excess funds, if any.
7. Voucher number and date of final payment.
8. A statement that all required contract administration actions have been fully and satisfactorily accomplished.
9. Name and signature of the contract administrator.
10. Date.

Material Management

Material Management

The planning, organization and control of all aspects of inventory embracing:

- Purchasing
- **Inventory**
- Warehousing
- Transportation
- Work-in-progress
- Shipping
- Distribution finished goods.

Inventory Management

Basic issues:

- What to order
- How much to order
- When to order
- Where to store

Fundamental Approaches to Managing Inventory

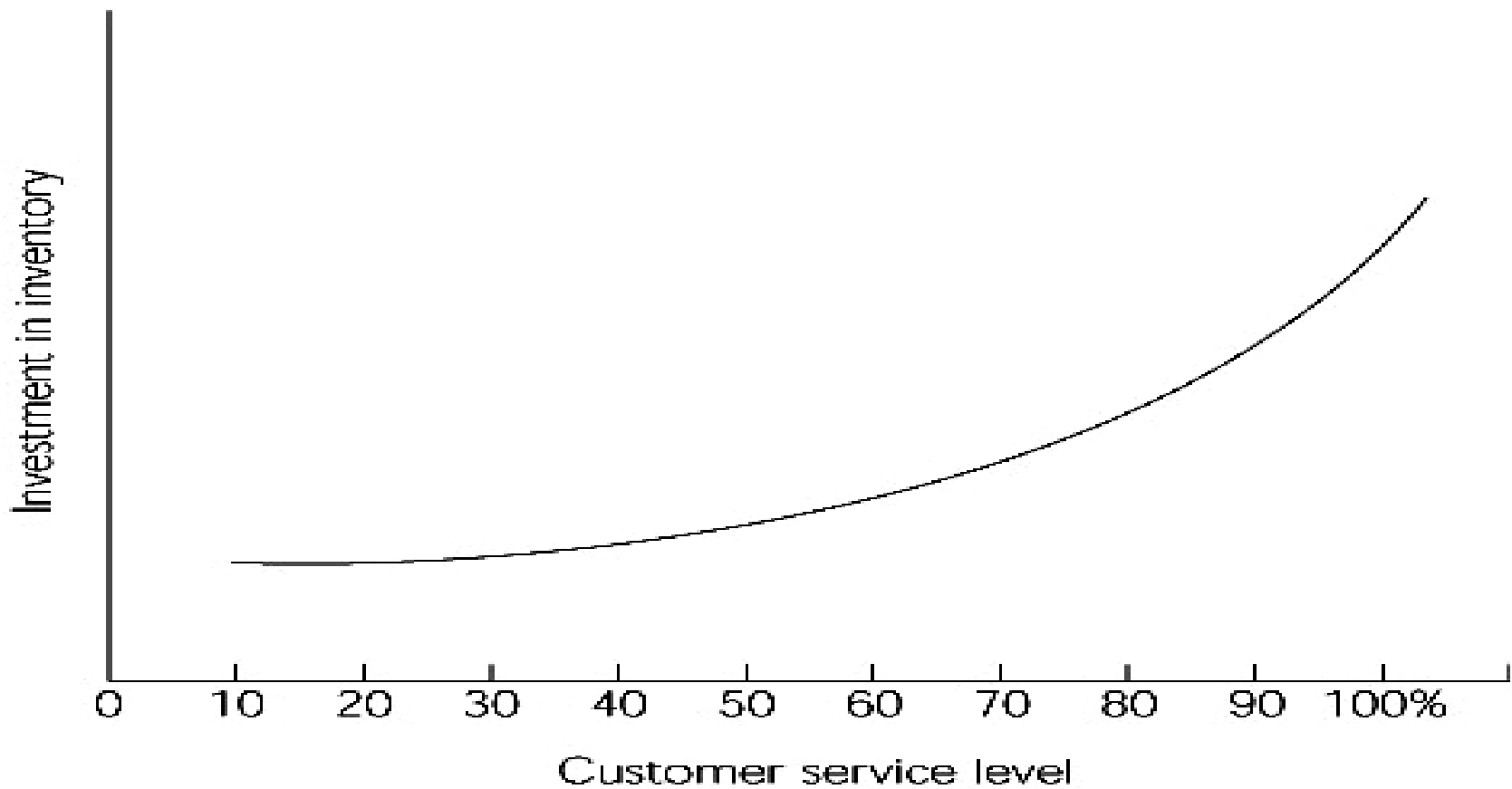
Traditionally, conflicts were usually present...

- customer service levels increased,
investment in inventory also increased

Emphasis now...

- increasing customer service and
reducing inventory investment

Relationship between Inventory and Customer Service Level



Managing Inventory

- Traditional Inventory
 - Economic Order Quantity (EOQ)
 - Re-order Point
 - Min-Max System
 - ABC Analysis
- Just In Time (JIT) or Kanban
- Vendor Manage Inventory (VMI/ Consignment)

JIT vs Traditional Inventory

- Reduces excess inventories
- Consistent lead times
- Quality
- Win-win relationships

VMI vs Traditional Inventory

- Shorter
- No inventories
- No waiting lines
- No lead times
- Consistent Quality
- Win-win relationships
- Company provide spaces

Drilling Material

Drilling Materials are items needed to carry out and complete a Drilling Project:

- Equipment
- Consumables
- Supplies

Drilling Material

Typical Drilling Materials:

- Rig
- Wellhead
- Christmas Tree
- Drill bits
- Casing & Tubing
- Drill pipe
- Etc.

Drilling Material

Key factors to considered

- Lead time
- Quality
- Supplier Reliability
- Logistics Transportation
- Warehouse
- Inventory

Drilling Material

Minimizing problems:

- Early Material/Equipment Planning
- Internal & External Integration
- Managing Inventory
- Logistics Transportation Planning
- Ensure Quality

Logistics Transportation

Definition:

A link that connects companies, customers, suppliers, warehouses, plants, and distribution centers

Arrange and facilitate to move materials or people from origin to destination, when and where needed

Logistics Transportation

Key elements:

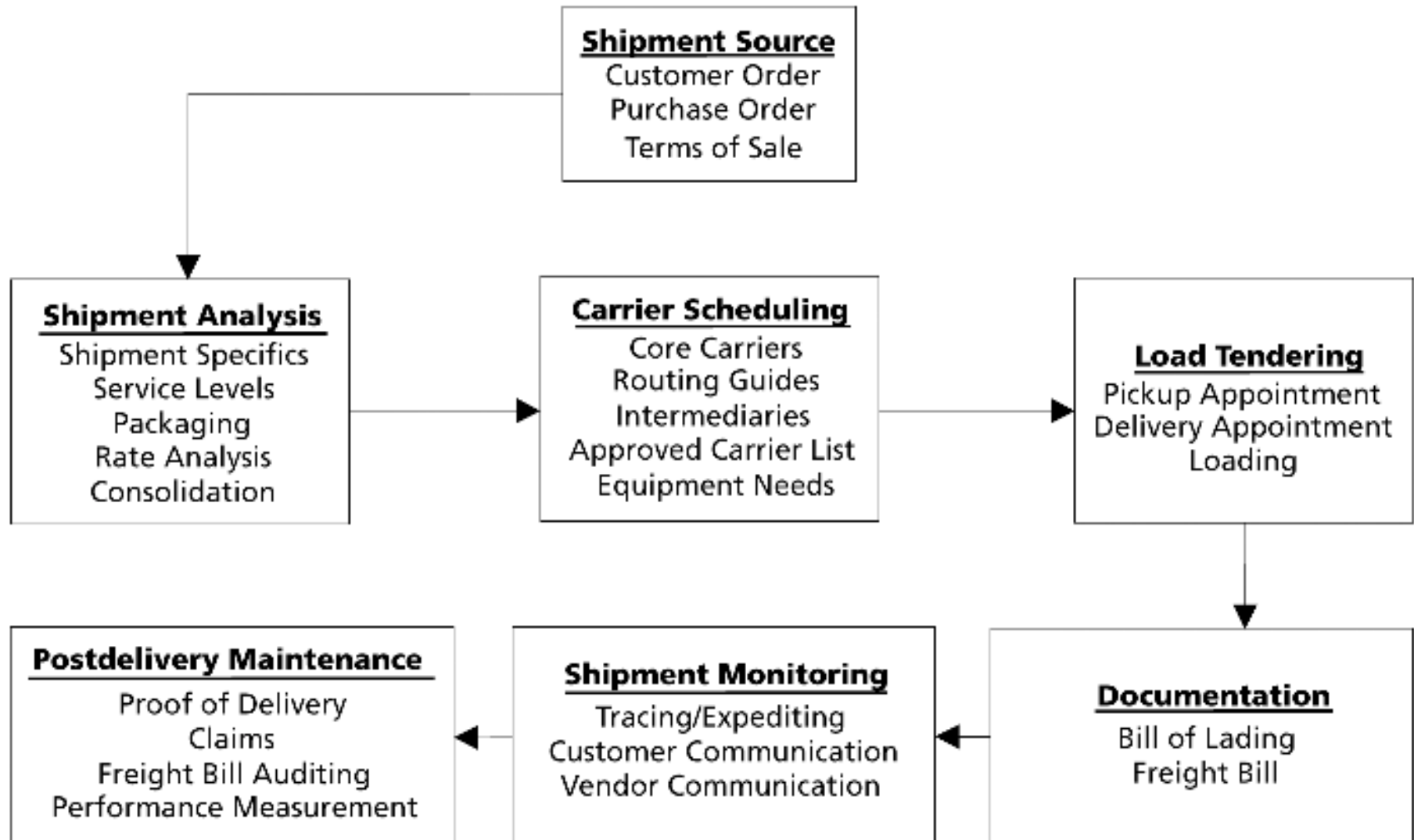
- Transportation Planning
- Transportation Documents
- Transit time
- Accessible to customers
- Service quality and timing
- Ability to move a variety of materials
- Ability to reach remote locations

Drilling Project Transportation

Items should be considered:

- Transportation schedule
- Transportation mode
- Transit time
- Provider quality
- Ability to move variety cargoes
- Permits & clearance

Transportation Management Process



End of Slides